

Django URLs Explained (Beginner-Friendly)

What are Django URLs?

Think of **Django URLs as the GPS or road map of your website.** 🗺️

When someone types a web address (URL) into their browser, Django decides **which Python function (called a view)** should handle that request and what page to display.

For example:

```
https://example.com/  
https://example.com/about/  
https://example.com/contact/
```

Each URL should display a different page. Django uses **URL routing** to connect each URL to the correct view.

Real-Life Analogy 🚗

Imagine a city:

- **Road** → URL
- **Traffic Police** → Django URL Dispatcher
- **Destination (House/Shop)** → Django View

If someone travels on the road:

```
/about/
```

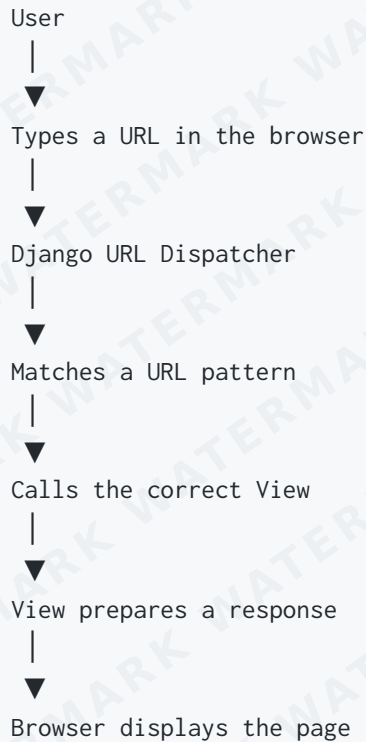
The traffic police directs them to the **About Page**.

Similarly, Django receives:

```
/about/
```

and sends the request to the appropriate Python function.

How Django URLs Work



The `urls.py` File

Every Django project contains a file named:

```
urls.py
```

This file stores all the URL routes for your application.

Example:

```
from django.urls import path
from . import views

urlpatterns = [
    path('', views.home),
    path('about/', views.about),
    path('contact/', views.contact),
]
```

Understanding `path()`

The basic syntax is:

```
path("URL/", view_function)
```

Example:

```
path("blog/", views.blog)
```

This means:

- If a user visits:

```
http://127.0.0.1:8000/blog/
```

- Django will execute:

```
views.blog()
```

Understanding Each URL

Home Page

```
path('', views.home)
```

URL:

```
http://127.0.0.1:8000/
```

Runs:

```
views.home()
```

About Page

```
path('about/', views.about)
```

URL:

```
http://127.0.0.1:8000/about/
```

Runs:

```
views.about()
```

Contact Page

```
path('contact/', views.contact)
```

URL:

```
http://127.0.0.1:8000/contact/
```

Runs:

```
views.contact()
```

What Happens if the URL Doesn't Exist?

Suppose a user visits:

```
http://127.0.0.1:8000/abc/
```

But your project does not contain:

```
path("abc/", ...)
```

Django cannot find a matching route, so it returns:

```
404 Page Not Found
```

Another Easy Analogy

Think of Django as a telephone operator.

Number Dialed Connected To

100 Police 101 Fire Department

Similarly:

URL Django Calls

`/about/` `about()` `/contact/` `contact()`

Summary

- A **URL** is the web address a user visits.
- Django stores URL routes inside `urls.py`.
- The **URL Dispatcher** checks every incoming request.
- If a matching route exists, Django calls the correct **View**.
- The **View** prepares a response (HTML, JSON, etc.).
- If no route matches, Django returns a **404 Page Not Found** error.

In simple words: Django URLs are like a map that tells Django **which Python function should run when someone visits a particular web address.**